3802 Lab 1, Part 2 Tips and Hints

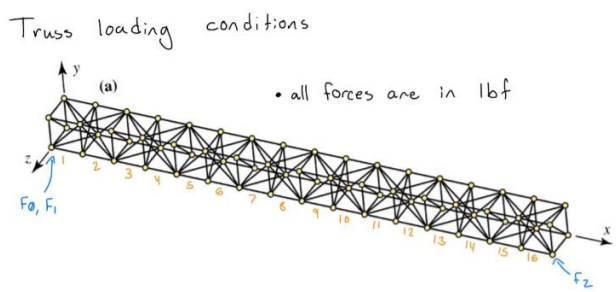
Part 2 Task 1: Analysis of Experimental Data

Provide plots of the reaction forces, displacements, and internal force measurements, for all cases presented. Plot each type of measurement for all cases in the same plot, in all cases as a function of the applied loading (i.e., plot the linear displacement for all cases in one plot).

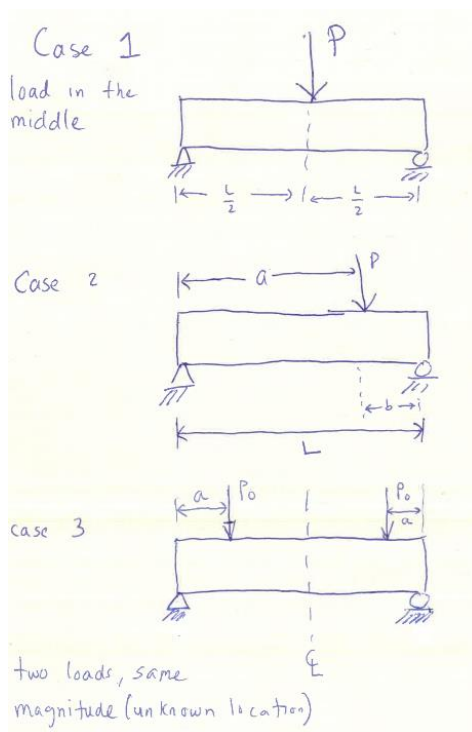
Compare the behavior of each loading case and discuss the results. Do they make sense?

Use linear regression and uncertainty analysis to verify the linearity of the measurements with respect to the external load magnitude. Explain the method of linear regression used and provide a plot of the regression curve. How do you confirm that the regression is linear?

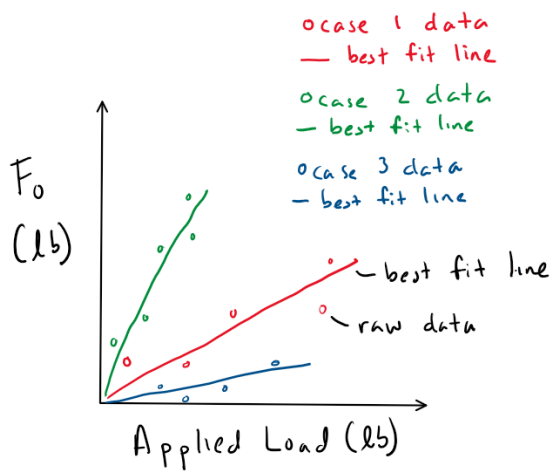
The reaction forces pertain to the load cells – there are three of these. Two load cells are placed on one end of the truss (F_0 and F_1), and the other is on the other end (F_2).



The 'cases' mentioned above are shown in the figure below.



One way to plot the results is to make the dependent variable ('y' axis) will be the reaction forces (F_0, F_1, F_2), the load cell at center of the truss (F3D), and the displacement from the LVDT. The independent variable ('x' axis) can be the applied load (these were the sandbags placed on the truss). In the example below, we see one of the reaction forces (F_0) plotted against the applied load. Three sets of data, and the trend line, are plotted, one for each case.



The above procedure can be repeated to generate a new plot for F_1 vs applied load, F_2 vs applied load, etc.

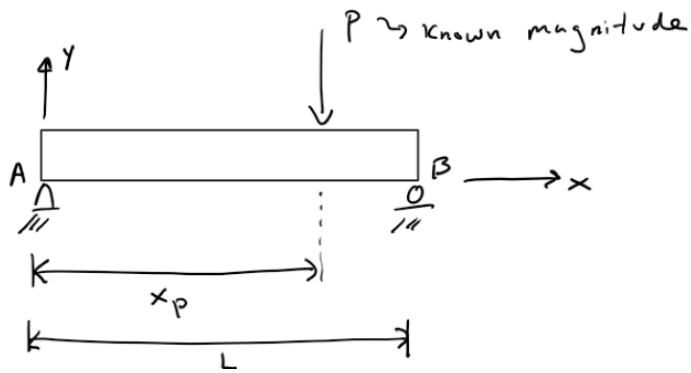
When discussing results, especially numerical values, **use a data table** to help summarize the findings. For example, the R^2 values can be placed into a table for easy viewing/discussion.

R^2	F_0	F_1	…
Case 1	0.912	0.876	…
Case 2	0.987	…	…
Case 3	0.787	…	…

Part 2 Task 2: Determine location of off-centered load case

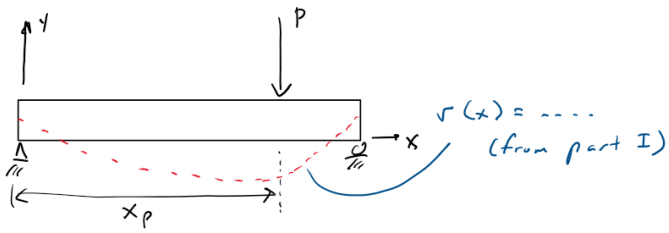
Consider second the case in which the load is applied at a single node, not in the center.

Can you identify the position of the loading, based on the reaction forces? Why? If not, use other information as needed, and explain the process. Show the position of the load. Compare the displacement and internal force from the experimental results and with the results from the equivalent beam model.

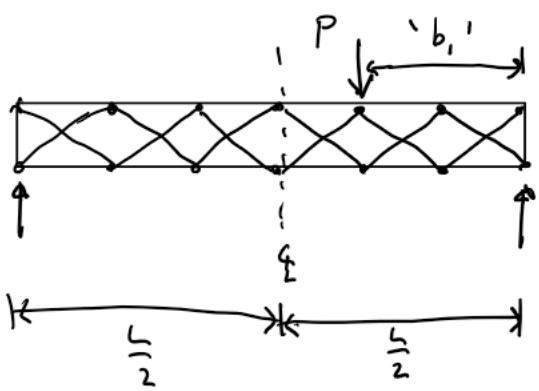


1) Draw FBD
2) use statics → find x_p in terms L, reaction forces, etc.
3) From the experimental data, the reaction forces are known.
4) use the data + your equation from step 2 to find x_p.

There is another way to do this! Beam deflection data can be utilized to estimate the position of the load. From part 1 you have an equation for beam deflection for a case where the load is placed at some random location on the beam. You also have experimental data for deflection (LVDT data). These two items can be used to estimate the deflection.



In your expression for v(x), set x = L/2. This expression now yields the vertical beam deflection at the mid-span (same as the LVDT). One can now run trial numerical experiments (basically iterations) to estimate where the load is located. Recall that the sandbags can only be placed on the joints/nodes (or where the circular rods come together). There are only a handful of possible load positions to consider.



1) guess a value for b_1

2) compute a deflection using $v(x=\frac{L}{2}) = \ldots$

3) compare that result to the LVDT data

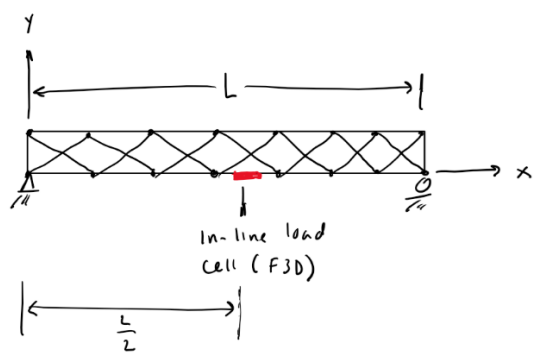
4) If the agreement is close, you have a good idea where the load is.

5) If the agreement is off, take another guess for b_1.

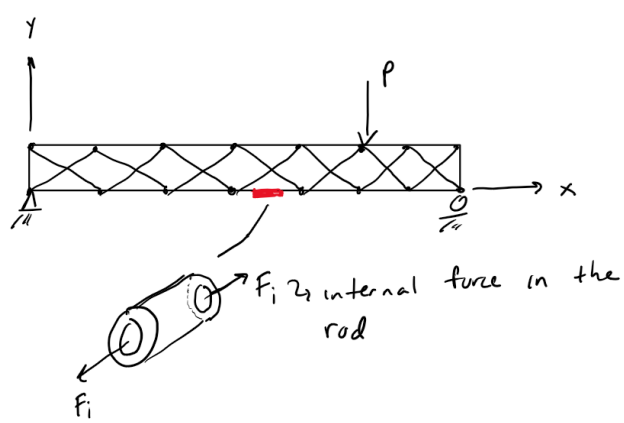
We are asked to compare the displacement from the experimental data to that of the equivalent beam model. One way to make this comparison is to look at the case where the load is at the mid-span. Using the beam deflection equation derived in part I, compute the deflection in the middle (x = L/2). **Summarize the results in a table.**

Mid-span beam deflection using equivalent beam model (in.)	LVDT data (in.)
…	…

We are asked to compare the internal force based on the equivalent beam model and the experimental data (F3D). The in-line load cell is placed near the mid-span of the truss. It measures the internal (axial) force of one of the hollow tubes, or rods.

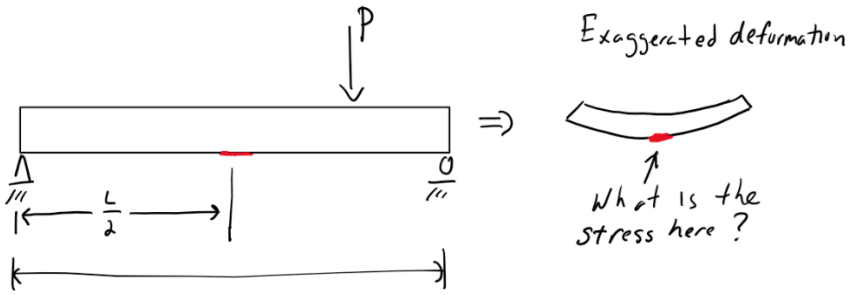


We can develop an expression to find F_i, which is the internal force in the rod shown below.



Recall that the rod, as drawn above, is in a state of tension and the stress in the rod is $\sigma = F_i/A$ where A is the cross-sectional area.

Another way to view this analysis is that the truss is treated as an 'equivalent' beam. When the load P is applied, **bending stresses** are developed in the beam. We can examine the bending stress near center (where the in-line load cell is located). Refer back to the 'Review of beam deflection and MoI' document used for Part I as bending stress was reviewed there if you are unfamiliar, or do not remember, how to compute the bending stress in a beam.



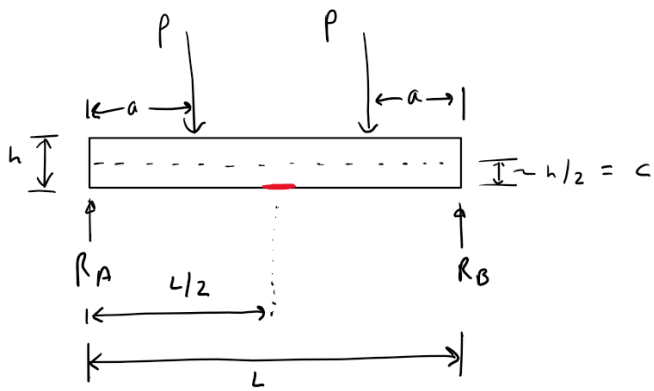
One can equate the stress from the axial loading to that of the bending and develop an expression determine F_i. This force, F_i, can be compared directly to the F3D (in-line) load cell data. For easy reading, you can **place the data in a table**.

Expected force (lb)	F3D data, mid-span (lb)
…	…

Part 2 Task 3: Determine the location of the symmetric two load case

Consider now the third case, in which the load is applied at two different nodes symmetrically, with the same load at each of them.

Can you identify the positions of the loads based on reaction forces? Why? If not, use other information as needed, and explain the process. Show the position of the loads. Compare the displacement and internal force from the experimental results and with the results from the equivalent beam model.



One can use a procedure like what was utilized in the prior task. Equate the stresses (axial and bending) at the mid-span, and this will allow one to develop an equation for the distance, 'a'. The expression for 'a' will be in terms of P, I, c, F_i, and rod-cross-section, A. The term F_i is the internal force (F3D data). When making use of the F3D here data one must account for the pre-load. Recall that there is small compression load applied to the load cell to hold in place on the truss (if the pre-load is lost, the load cell would just fall out).